

Sustainable Bio-Waste Management: Human Hair Based Heavy Metal Removal

Smart Water Quality Monitoring and Automated Emergency Alerting
System

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The Critical Challenge

Industrial Contamination: Factories continuously discharge extremely dangerous heavy metals like Lead (\$Pb\$) and Mercury (\$Hg\$) into fresh water sources. These chemicals are non-biodegradable, poisoning entire aquatic ecosystems.

Health & Cost Crisis: These toxins bioaccumulate in human organs causing irreversible damage. Standard water purification machinery is highly expensive and impossible to afford for small-scale local businesses.

Bio-waste Overflow: At the same time, tons of human hair are dumped from hair salons daily, blocking urban drainage networks and overloading municipal landfills.



The Power of Keratin



Waste Into Wealth (Circular Economy)

- 🌱 **Natural Biosorption:** Human hair contains rich quantities of ****Keratin****, a high-affinity fibrous structural protein.
- 🔄 **Chelation Process:** Functional active groups present inside Keratin organically capture and chemically bind heavy metal ions.
- 🌍 **Low Cost & Green:** Recycles a zero-value salon solid waste into a highly efficient chemical-free filtration medium.

Project Objectives



Reuse Bio-Waste

Clean and repurpose unutilized local salon hair waste to address the critical landfill accumulation issues in cities.



IoT Water Check

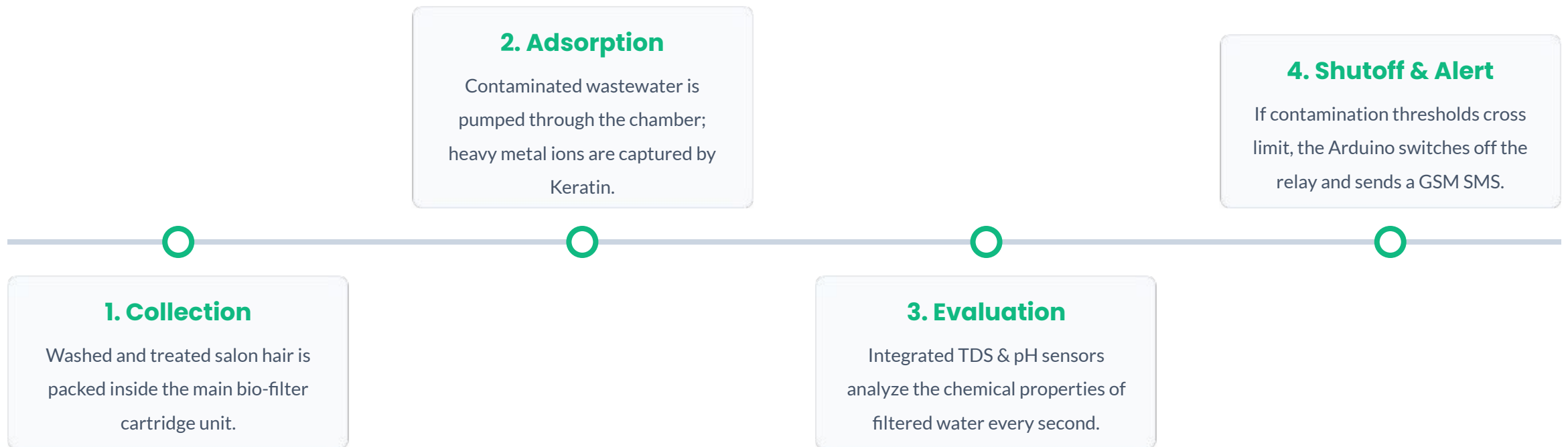
Integrate robust TDS and pH sensors with an Arduino microcontroller for 24/7 continuous automatic water evaluation.



Instant Alerts

Develop a remote GSM-based cellular emergency subsystem to send instant notification alerts to smartphones during failures.

How It Works



System Architecture

Subsystem Component	Type / Specification	Core Operational Responsibility
Bio-Filter Cartridge	Treated Salon Human Hair	Naturally binds with toxic Mercury and Lead ions (\$Pb\$, \$Hg\$).
Sensors Subsystem	TDS & pH Sensors	Continuous real-time checking of water purity and acidity levels.
Arduino Uno	ATmega328P MCU	Central logic unit executing automated control algorithms.
Actuator Control	5V Relay Switch + Pump	Closes or instantly cuts off water flow depending on purity state.
GSM module	SIM900 / SIM800L	Transmits remote SMS warning messages to user's smartphones.

| Advantages & Impact

100%

Eco-Friendly & Sustainable

Key Project Strengths:

- ✓ **Zero-Cost Filtration:** Uses easily sourced waste hair as active raw material.
- ✓ **High Surface-to-Volume Ratio:** Natural fiber structure achieves much faster and more comprehensive filtration.
- ✓ **Highly Portable:** Can be set up in disaster regions to quickly provide safe utility water.
- ✓ **Saves Local Drainage:** Resolves municipal salon solid waste clogging.

Simulation & Roadmap

Live Simulation Output

Tested on Wokwi Platform: Programmed with responsive conditional logic to prevent any accidental toxic outflows.

Standard State: Water passes through normally (TDS: 345 ppm | pH: 7.2) -> Pump is ON.

Contaminated State: When high TDS (>1000 ppm) is simulated, the Relay triggers, switches the pump OFF and serial console alerts 'Status: Heavy Metals Detected!'

Future Research Roadmap

Multi-layer Cartridges: Expanding with a hybrid layout structure to target active synthetic pollutants alongside heavy metals.

Chemical Desorption: Engineering simple wash extraction methods to safely extract the trapped metals and reuse the hair core.

AI Analytics Integration: Integrating smart ML regression curves to estimate remaining filter life based on real-time sensor history.

Image Sources



<https://daikiaxis.lk/wp-content/uploads/2023/07/Industrial-Water-Pollution-and-Its-Effects.jpg>

Source: daikiaxis.lk



<https://media.defense.gov/2010/Jul/14/2000343433/2000/2000/0/100714-F-0000X-365.JPG>

Source: www.af.mil